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REVIEW



Does oxidative stress contribute to toxicity in acute organophosphorus poisoning? – a systematic review of the evidence

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ABSTRACT

Introduction: Organophosphorus (OP) insecticide self-poisoning is a global problem, killing tens of thousands of people every year. Oxidative stress has been proposed to play a pathological role in OP poisoning, but whether it plays a direct toxic role is currently unclear.

Objectives: To determine whether there is consistent evidence of oxidative stress in patients with acute OP insecticide self-poisoning, and whether there are animal or human trial data that indicate that treatment of oxidative stress provides clinical benefit, which would suggest a direct toxic effect of oxidative stress.

Methods: We conducted a systematic review using the PubMed, EMBASE and MEDLINE databases, and the Cochrane Database of Systematic reviews, based upon the following search terms and keywords: “organophosphate poisoning”, “oxidative stress” and “antioxidant”. All articles relevant to the aims of the study were included. Articles related to chronic OP poisoning, to use of medicines without antioxidant benefits, or to subjects other than oxidative stress were excluded. The search returned 256 results of which 17 studies were considered relevant and grouped under the following categories: observational human studies ($n=11$) and intervention studies in animals ($n=4$) and humans ($n=2$).

Oxidative stress markers in human studies: Oxidative damage to lipids and proteins was reported in all eleven human studies. Eight of nine studies reported variable increases in a weak marker of lipid peroxidation, malondialdehyde. In two case-control studies, erythrocyte membrane malondialdehyde concentrations were 380% and 160% higher in cases than controls, while plasma malondialdehyde concentrations were ~63% higher in cases than controls in three case-control studies. In a prospective study, plasma malondialdehyde did not increase significantly from baseline in moderate or severely poisoned patients. Five case-control studies measured thiol residues as markers of protein oxidative damage and found variable changes after poisoning. No evidence of oxidative DNA damage was found in the one study that investigated it.

Antioxidant intervention studies in animals: After treatment with an antioxidant, all four studies showed an improvement in either markers of oxidative damage or antioxidant activity. One mouse study with a relatively low risk of bias showed that administration of acetylcysteine 200 mg/kg reduced malondialdehyde by 35% and increased survival by more than 60%.

Antioxidant intervention studies in humans: We found two small randomised controlled trials reporting the use of acetylcysteine as an adjunct to standard treatment in acute OP poisoning. The trials found that acetylcysteine reduced atropine requirements by 77% and 55%, but did not affect clinically relevant outcomes.

Conclusions: Several studies showed evidence of OP insecticide-induced oxidative damage and antioxidant activity, suggesting that endogenous antioxidant defences are triggered in acute OP poisoning. However, the markers of lipid peroxidation used were weak, there was high inter-individual variability between studies in results and quality, and marked variation between the OP insecticides involved. Animal data provide some evidence that antioxidants alleviate adverse effects of acute poisoning, suggesting that oxidative stress may directly cause clinical harm. Acetylcysteine appeared beneficial in animal studies, but this could be mediated *via* increased synthesis of the endogenous detoxifying agent, glutathione, rather than through a direct antioxidant effect. The two human clinical studies were too small to provide any clear evidence to support the use of acetylcysteine in OP poisoning. Further research into the mechanisms of oxidative stress in acute OP poisoning, combined with large unambiguous clinical trials of antioxidants, are required before they can be used routinely in treatment.

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Introduction

Acute organophosphorus (OP) insecticide self-poisoning is estimated to result in around one hundred thousand deaths every year, predominantly in rural Asia [1,2]. OP compounds cause acetylcholinesterase inhibition that results in overstimulation of cholinergic receptors and the acute cholinergic syndrome. Affected individuals develop acute respiratory failure that must be treated with mechanical ventilation [2], while some develop a delayed neuromuscular junction dysfunction, resulting from persistent activation of neuromuscular junction-associated nicotinic acetylcholine receptors. The resultant paralysis – known as type II paralysis or intermediate syndrome – requires prolonged ventilatory support [3,4]. The core treatment for acute OP poisoning includes supportive care, oxygen, atropine and oximes [5]. Although the existing treatment regimen is effective at reversing some of the effects of acetylcholinesterase inhibition, it is not always effective and case fatality remains high.

Some researchers have hypothesised that oxidative stress is a mechanism behind the complications associated with OP insecticide poisoning [6–10]. The suggestion is that mitochondrial dysfunction, ischaemia reperfusion in muscle tissues, or inactivation of endogenous antioxidants, are mechanisms responsible for causing oxidative damage in OP poisoning, with knock-on damaging effects [6,9,10].

Oxidative stress

Oxidative stress (Box 1) is defined as an imbalance between oxidants and antioxidants in favour of oxidants that leads to a disruption of reduction-oxidation (redox) signalling and/or molecular damage [11]. Antioxidants constitute a battery of endogenous defences which rapidly inactivate oxidants that have the potential to cause harm – collectively known as reactive oxygen species (e.g., superoxide ($\text{O}_2^{\cdot-}$), peroxy ($\text{ROO}^{\cdot}$) and hydroxyl ($\text{OH}^{\cdot}$) free radicals, and hydrogen peroxide (H_2O_2)) [12]. The term “nitrosative stress” is sometimes used to refer to the oxidative damage caused by reactive nitrogen species (e.g., nitric oxide (NO) and peroxynitrite (ONOO^-)).

Reactive oxygen species and reactive nitrogen species free radicals (Figure 1) that overcome antioxidant defences can damage nucleic acids, lipids, free amino acids and carbohydrates; the resultant damage can progress to mutagenesis, apoptosis or necrosis [13,14]. Different free radicals have variable ranges of reactivity and are associated with different antioxidant systems [11]. Hydroxyl radical ($\text{OH}^{\cdot}$) is the most reactive oxidant and hydrogen peroxide (H_2O_2) is one of the least reactive oxidants [11,15]. The reactivity of an oxidant determines its *in vivo* lifetime and the distance it can travel before reacting with a cellular component – the more reactive an oxidant is, the shorter its half-life and the shorter the distance it can travel within the cell [16].

BOX 1 . Oxidative stress in healthy and pathological states.

Dysfunction in the system on account of inappropriate concentrations of substrate or oxygen, or through poisoning of components of the respiratory chain, can result in incomplete reduction of molecular

oxygen to water, generating the powerful reactive oxygen species, superoxide [46]. Dysfunctional respiration is an important source of potentially harmful oxidants, along with specific enzymes (e.g., cyclo-oxygenases, NAD(P)H oxidases, and xanthine oxidase) [12,46].

In health, reactive oxygen species generation is typically very low and has an important signalling function [11]. Excess reactive oxygen species is rapidly inactivated by a wide range of antioxidants present in both cellular and extracellular compartments. However, excess generation of reactive oxygen species and/or dysfunctional antioxidant defence can result in exposure of sensitive cellular components (DNA, proteins, lipids) to ROS-induced damage and consequent cellular damage – this is known as oxidative stress (Figure 1) [12,47,48].

Oxidative stress can be both a cause and a consequence of pathology [17]. The highly reactive nature of reactive oxygen species makes their direct measurement problematic, particularly *in vivo*. Surrogate markers are therefore the mainstay of measurement of reactive oxygen species. Oxidative damage to proteins, lipids or nucleic acids [18] has traditionally dominated the field, but depletion of antioxidants and adaptive responses to oxidative stress mediated by upregulation of antioxidant enzymes is also a legitimate approach. However, it is important to understand that all measures of oxidative stress have limitations and none in isolation will provide a complete picture (Box 2).

BOX 2 . Measuring oxidative stress.

Measuring the damage caused to macromolecules is generally more reliable than measuring reactive species because reactive oxygen species/reactive nitrogen species have very short half-lives. Therefore, despite the availability of sophisticated methods to directly measure reactive species *in vitro*, they are often not applied clinically or in *in vivo* studies [49]. Measuring the abundance of antioxidants in accessible compartments (usually blood cells or blood plasma) is one approach that can be adopted to estimate the (im)balance of oxidants/antioxidants *in vivo*. However, the relevance of such measures in blood plasma is limited because this compartment is distant from potential sources of oxidants and is only loosely related to intracellular antioxidant status. Measurements in red blood cells or inflammatory cells are more appropriate, but it should be recognised that antioxidant status is a complex and dynamic state. For example, oxidative stress is well known to induce an adaptive up-regulation of antioxidant enzymes through the NRF2/ARE-1 pathway, triggering a counter-intuitive increase in intracellular enzymes (e.g., superoxide dismutase, glutathione peroxidase) or sacrificial antioxidants (e.g., glutathione *via* up-regulation of the rate-limiting enzyme in its synthesis, glutamyl-cysteine ligase) [50].

A wide range of techniques are available to measure these antioxidant systems, from colorimetric assay kits to chromatographic methods and even mass spectrometry; the latter are regarded to be more specific, but colorimetric methods are often preferred on the basis of convenience and cost. The alternative to measuring antioxidant status is to measure surrogate markers of oxidative stress in the form of oxidised macromolecules. Commonly used methods to measure damage caused by oxidative stress are often specific to the particular macromolecule damaged. Table 1 summarises the changes expected in biomarkers of oxidative stress under conditions of oxidative stress.

Oxidative damage to lipids (lipid peroxidation) is widely studied due to the abundance of lipids in the body. Until recently, one of the most popular quantitative biomarkers of lipid peroxidation has been malondialdehyde, often measured using a classical chemical method, the thiobarbituric acid (TBARS) assay. However, the TBARS method of measuring malondialdehyde has been heavily criticised for its lack of specificity. As a result, other measures of malondialdehyde have been developed and are considered more reliable, including fluorescence and high performance liquid chromatography (HPLC) [49]. Irrespective

of the method used to measure malondialdehyde, however, this marker is generally regarded to be a poor marker of oxidative stress.

Alternative markers of lipid peroxidation are 4-hydroxynonenal (measured by HPLC or more sensitively by gas chromatography mass spectrometry; GC-MS) [49] and F₂-isoprostanes (measured in urine or plasma by ELISA or, preferably, by mass spectrometry [LC or GC-MS]) [38]. F₂-isoprostanes and 4-hydroxynonenal are considered to be more specific indicators of lipid peroxidation than malondialdehyde because they are more stable molecules and the TBARs assay for malondialdehyde is influenced by its reaction with several other compounds found in plasma [37,38,49]. Malondialdehyde concentrations rise after acute and chronic inflammation, or after exposure to toxins, suggesting that it can be a good marker of toxicity or inflammation, but it lacks specificity for oxidative stress per se [51,52]. 4-hydroxynonenal is a product of lipid peroxidation that modifies protein. An increase in hydroxynonenal leads to an increase in hydroxynonenal-modified GAPDH which eventually undergoes ubiquitination. Conjugated dienes are chemical structures with two double bonds separated by a single bond. They are used as evidence for lipid peroxidation as they are not found in unsaturated fatty acids; rearrangement of the double bond to the conjugated diene formation is mediated by reaction with oxygen or ROS [53]. Dienes are usually measured spectrophotometrically by their absorbance at 234 nm. Although accepted as "evidence for lipid peroxidation", dienes are not used for "absolute quantification" because a wide range of other compounds also absorb at this wavelength [53].

Oxidative damage to proteins: Oxidised proteins are chemically stable and can be used as indicators of whole-body oxidative stress [38]. Thiol groups are organosulphur compounds that represent a substantial antioxidant pool in mammalian systems. Thiols can be classified into protein and non-protein thiols [54]: glutathione is the most abundant non-protein thiol, followed by cysteine, and other low molecular weight thiols [55]. Low concentrations of reduced glutathione or a high ratio of oxidised glutathione (glutathione disulphide) to glutathione in blood or cells are indicators of oxidative stress. Glutathione and glutathione disulphide can be measured using classical chemistry and spectrophotometry, HPLC or GC-MS [49].

Protein thiols are usually present in higher concentrations than cysteine or glutathione due to abundance of albumin [55]. They are found to be decreased under oxidative stress as they transform into mixed disulphides on account of oxidation-mediated reaction with low molecular weight thiols; decreased concentrations of albumin in some disease conditions might also be a contributing factor [56]. Depending on the pathology, specific thiol groups have been found to increase under oxidative stress. Typically, however, the concentration of total reduced thiols decreases in the face of oxidative stress, probably due to sacrificial oxidation of glutathione [56].

Oxidation of cysteine residues under oxidative stress leads to S-thiolation, i.e., the formation of mixed disulphides between low-molecular mass thiols and protein thiol groups, such as glutathione (S-glutathionylation) [57]. As glutathione disulphide accumulates in cells, disulphides get exchanged with protein thiols leading to S-glutathionylation of the cysteine residues in the proteins. S-glutathionylated proteins increase under oxidative stress in many pathological conditions [57].

3-nitrotyrosine (NO₂-Tyr), a biomarker for "NO-derived oxidants", can be measured by ELISA or (more specifically) by HPLC or LC-MS.

Of note, protein carbonyls are the preferred marker of protein oxidation because plasma concentrations are stable and can be measured through HPLC, immunoblot, ELISA, fluorometric and spectrophotometric assays [18,38]. Protein carbonyls were not assayed in any of the studies reported here.

Oxidative damage to nucleic acids: products of oxidative DNA damage such as thymidine glycol and 8-hydroxydeoxyguanosine can be measured through HPLC, GC-MS, LC-MS or ELISA methods [49]. 8-hydroxydeoxyguanosine is present in urine samples and can be used to assess whole-body oxidative stress [49].

Objectives

To determine:

1. whether there is consistent human evidence of oxidative stress in acute OP insecticide self-poisoning clinically, or at a cellular level;
2. whether there is animal or human evidence of clinical benefit resulting from antioxidant treatment, which would suggest a causative role for oxidative stress.

Methods

PubMed, MEDLINE and EMBASE databases were searched using the MESH terms "organophosphate poisoning", "oxidative stress" and "antioxidant" as either a subject heading or a key term. The Cochrane Database of Systematic Reviews searches were performed using the term "organophosphate poisoning" as the subject heading. Studies were also chosen from our personal library, based on relevance, and the bibliographies of papers were hand-searched. There was no language restriction. Studies were excluded if they met any of the following criteria: (1) articles related to chronic rather than acute OP poisoning; (2) animal studies that did not address mechanisms of oxidative stress; and (3) clinical studies that considered medicines without any known antioxidant effects according to DrugBank [19].

Both authors involved in the selection process (LR, ME) independently selected papers for inclusion. All relevant articles, full-text journal articles, articles ahead of print, in-process and non-indexed citations were included. Conference abstracts were excluded. Disagreements were resolved by discussion.

Data analysis

We conducted a narrative synthesis of all the relevant articles. The validity of studies was determined based on sample size, the methodology used in the study, and interpretation of researchers' findings. In particular, we focused on assessing the reliability of methods used by researchers to measure oxidative stress (Box 2). Factors such as when blood samples were taken, potency of the poison ingested, severity of poisoning, and treatment protocol were also analysed for reliability.

The antioxidant intervention studies in animals were assessed for bias using the Systematic Review Centre for Laboratory Animal Experimentation (SYRCLE)'s risk of bias tool that is an adapted version of the Cochrane risk of bias tool [20]. The clinical intervention studies were assessed for risk of bias using the Cochrane Collaboration guidelines [21].

Results

We found a total of 17 studies that were relevant (Figure 2) to the aim of our review. The studies were classified under the following categories: human observational clinical studies of acute self-poisoning ($n = 11$), antioxidant intervention studies in animal models ($n = 4$), and antioxidant intervention studies in OP insecticide poisoned patients ($n = 2$) [7,8,22–36].

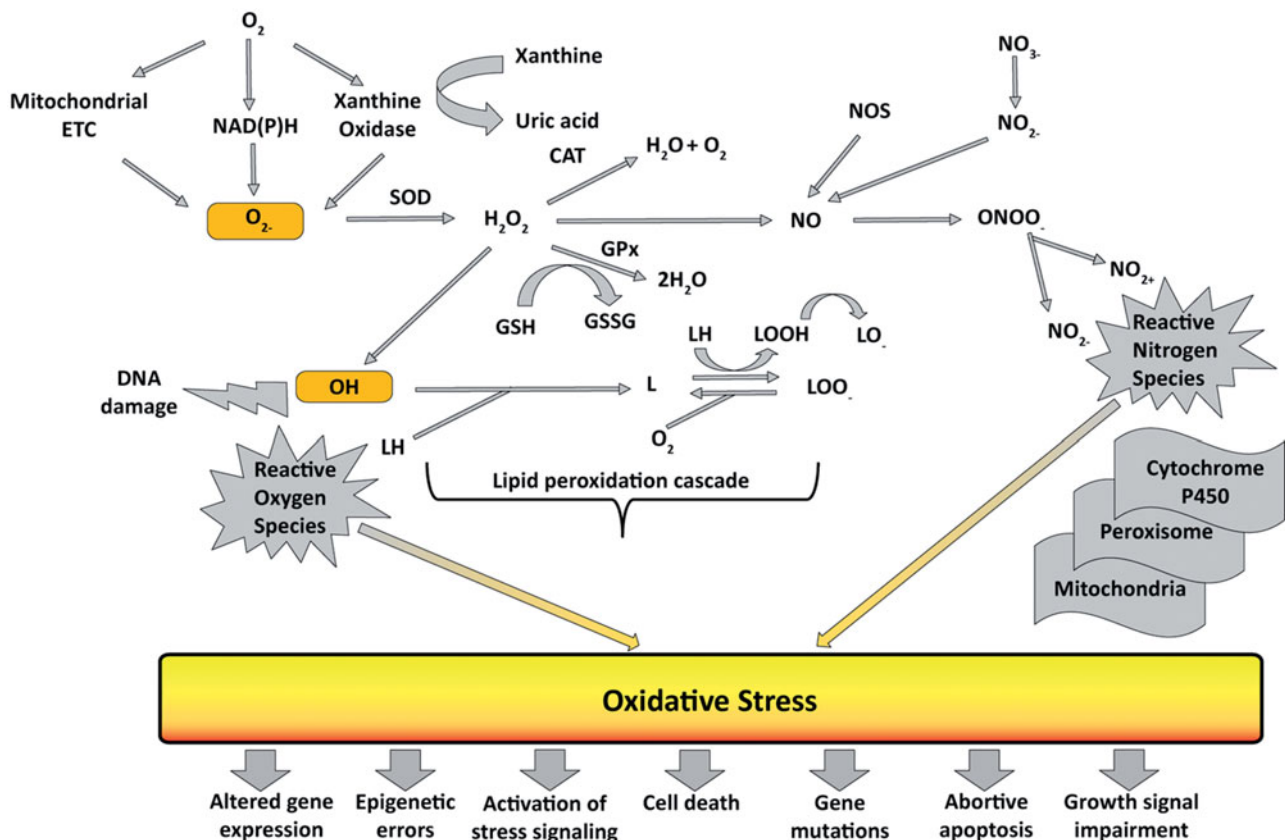


Figure 1. Pathway showing reactive oxygen species (ROS)/reactive nitrogen species (RNS) system – Highly reactive free radical (or oxidant) superoxide (O_2^-) released from mitochondrial electron transport chain (ETC), xanthine oxidase, or other source, is converted to a less reactive and more stable compound hydrogen peroxide (H_2O_2), by antioxidant superoxide dismutase (SOD). Other antioxidants such as catalase (CAT) and glutathione (GSH) also counteract free radicals. However, if free radicals are present in excess, it could result in oxidative stress which causes disruption to various cellular processes as shown. This figure was adapted with permission from figure “pathway showing ROS/RNS system” from the paper “Oxidative stress: major executioner in disease pathology, role in sperm DNA damage and preventive strategies” [58].

Table 1. The behaviour of (quantitatively measured) oxidative stress markers under oxidative stress.

Marker	Under oxidative stress
<i>Lipid peroxidation</i>	
Malondialdehyde	↑
4-hydroxynonenal	↑
Conjugated dienes	↑
F ₂ -isoprostanes	↑
<i>Oxidative damage to proteins</i>	
Reduced glutathione; glutathione: glutathione disulphide ratio	↓
Protein carbonyls	↑
2-nitrotyrosine	↑
Protein thiols	↓
Total thiol groups	↓
<i>Oxidative damage to DNA</i>	
8-hydroxydeoxyguanosine	↑
<i>Antioxidant and antioxidant-dependent enzymes</i>	
Superoxide dismutase	↑/↓
Catalase	↑/↓
Glutathione peroxidase	↑/↓
Glutathione reductase	↑/↓

Oxidative stress markers in human studies

We found 10 case-control studies and one prospective observational study including a total of 579–584 patients (one paper provided a patient number range) [26–36]. The diagnosis of acute OP insecticide self-poisoning was made

clinically, based on history, bottle brought with the patient, presence of characteristic symptoms, and investigation findings. In nine studies, samples were taken during admission. One study does not specify when the samples were collected. Prospective and case-control studies reported on blood samples from the poisoned patients and oxidative stress markers were usually compared with the activity of acetylcholinesterase or butyrylcholinesterase enzymes (both inhibited by OP compounds).

Oxidative damage

Lipid damage

Nine studies (eight case-control, one prospective observational study) measured lipid peroxidation using malondialdehyde as a marker of oxidative stress (Box 2). Classical chemical and spectrophotometric methods were the two most common ways of measuring malondialdehyde. No study used high performance liquid chromatography, despite its higher specificity [37].

The eight case-control studies all found that mean erythrocyte, plasma or serum malondialdehyde was on average higher (mean increase: 115%, range: 32% to 379%) in acutely poisoned patients prior to treatment when compared to controls (Table 2) [26,28–30,32,34–36]. In one case-control study that measured plasma malondialdehyde by a classical

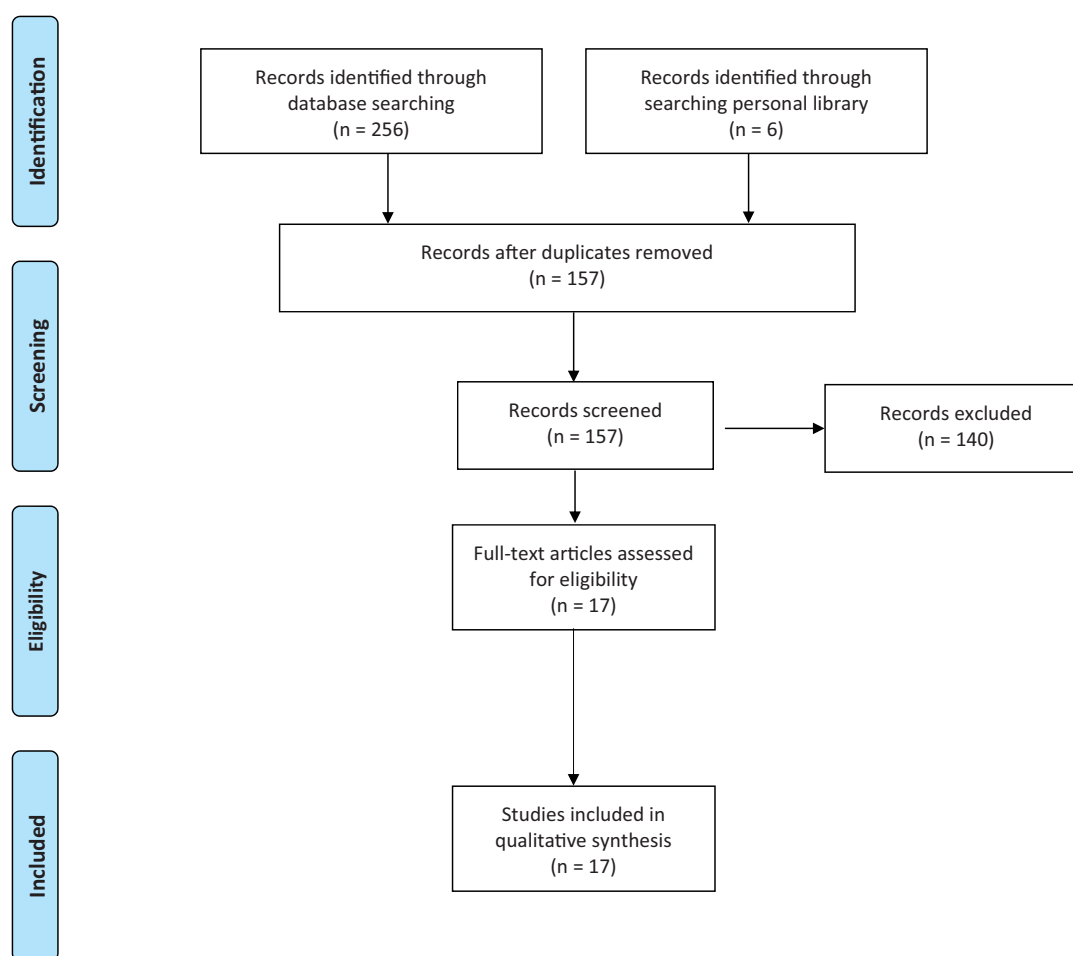


Figure 2. PRISMA flowchart for the systematic review.

chemistry method, mean plasma malondialdehyde was twice as high (+102.9%) in severely poisoned patients than controls [36]. However, a case-control study of plasma malondialdehyde using spectrophotometry found plasma malondialdehyde to be increased by a much smaller amount (32.1%) compared to controls [34]. In the prospective study, plasma malondialdehyde increased non-significantly by 5% and 7% from baseline in moderate and severely poisoned patients, respectively [27].

Erythrocyte malondialdehyde was higher in acutely poisoned patients than plasma or serum malondialdehyde (Figure 3). In two case-control studies, malondialdehyde concentrations in erythrocyte membranes were raised by 379% (control group: 0.56 ± 0.405 U/mg protein; patient group: 2.68 ± 4.74 U/mg protein) at admission and 162% (control group: 12.7 ± 2.45 nmol TEP/mg protein; patient group: 33.24 ± 27.65 nmol TEP/mg protein) during the course of their hospital stay compared to control groups (Table 2) [28,30]. One study reported a 202% mean increase in the erythrocyte malondialdehyde concentration in patients with intermediate syndrome [30]. Two case-control studies reported a positive correlation between malondialdehyde concentration and clinical severity [26,35].

Two studies in patients found that conjugated diene concentrations were significantly raised ($p < 0.01$) by 312%

(control group: 19.04 ± 22.7 U/mg protein; patient group: 78.47 ± 144 U/mg protein) at admission and 374% (control group: 0.23 ± 0.06 U/mg protein; patient group: 1.09 ± 0.91 U/mg protein) during the course of their hospital stay compared to control groups [28,30].

Protein damage

We found four case-control studies that measured thiol residues specific to protein oxidation [28,30,33,36,38].

Two small studies found a depression of chemically reduced thiol groups on admission in acutely poisoned patients vs. controls, suggesting that there was significant oxidation of thiol residues in the patient group (Figure 3) [33,36]. Two studies utilising wet chemistry methods failed to show a significant increase in thiol groups bound to protein in acutely poisoned patients at admission [28,30]. However, a significant increase in protein thiol group concentration was detected on day 7 in one study of intermediate syndrome patients compared to non-intermediate syndrome patients, which then doubled by day 10 [28]. Another case-control study concluded that intermediate syndrome patients had greater oxidative damage to proteins due to a mean increase of 19% (intermediate syndrome: 0.25 ± 1.13 U/mg protein; non-intermediate syndrome: 0.21 ± 0.13 U/mg protein) in protein thiol groups in intermediate syndrome patients compared to non-

Table 2. Human clinical observational studies.

Authors, year	Study type	Sample collected ^a	Number of samples	Method	Results
<i>Malondialdehyde (MDA, lipid peroxidation)</i> Banerjee et al., 1999 [29]	Case-control study	Serum	25–30 patients; unknown number of healthy controls	Classical chemistry	Control group: 2.95 ± 0.85 nmol/ml Patient group: 4 ± 0.95 nmol/ml Percentage difference: 35% mean increase from control
Zhou et al., 2002 [34]	Case-control study	Plasma and erythrocyte	58 patients; 58 healthy controls	Spectrophotometric method	Control group: Plasma: 11.04 ± 1.6 μ mol/L Erythrocyte: 28.72 ± 4.13 nmol/g Hb Patient group: Plasma: 14.59 ± 2.12 μ mol/L Erythrocyte: 39.39 ± 5.53 nmol/g Hb Percentage difference: Plasma: 32.2% mean increase Erythrocyte: 37.1% mean increase from control
Dandapani et al., 2003 [30] ^b	Case-control study	Erythrocyte	19 patients; 5 healthy controls	Classical chemistry	Control group: 12.7 ± 2.45 nmol TEP/mg protein Patient group: 33.24 ± 27.65 nmol TEP/mg protein (not statistically significant compared to control) Percentage difference: 161.7% mean increase
Vidyaasagar et al., 2004 [35]	Case-control study	Plasma	84 patients; 10 healthy controls	Classical chemistry	Control group: 3.6 ± 0.9 nmol/ml Patient group: Grade 1: 5.3 ± 1.5 nmol/ml Grade 2: 5.9 ± 2 nmol/ml Grade 3: 5.7 ± 1.8 nmol/ml Grade 4: 6 ± 1.9 nmol/ml Grade 5: 6.7 ± 2.3 nmol/ml Percentage difference: 53.7% mean increase Increase in severity of poisoning from grade 1 to grade 5, increased MDA levels (47.2% mean increase at grade 1 to 86.1% mean increase at grade 5)
Ranjbar et al., 2005 [36]	Case-control study	Plasma	22 patients; unknown number of healthy controls	Classical chemistry	Control group: 4.1 ± 0.37 nmol/ml Patient group: 8.32 ± 0.75 nmol/ml Percentage difference: 102.9% mean increase
Venkatesh et al., 2006 [28]	Case-control study	Erythrocyte	32 patients; 18 healthy controls	Classical chemistry	Control group: 0.56 ± 0.405 U/mg protein Patient group: 2.68 ± 4.74 U/mg protein Percentage difference: 378.6% mean increase
Zhang et al. 2010 [32]	Case-control study	Serum	49 patients; 50 controls	Spectrophotometric method	Control group: 3.54 ± 1.27 nmol/ml Patient group: 6.63 ± 1.91 nmol/ml Percentage difference: 87.3% mean increase
Hundekari et al., 2013 [26]	Case-control study	Serum	150 patients; 30 healthy controls	Classical chemistry	Control group: 1.05 ± 0.37 nmol/ml Patient group: Grade 1: 1.97 ± 0.39 nmol/ml Grade 2: 2.13 ± 0.55 nmol/ml Grade 3: 2.16 ± 0.52 nmol/ml Grade 4: 2.39 ± 0.61 nmol/ml Grade 5: 2.52 ± 0.64 nmol/ml Percentage difference: 94% mean increase Increase in severity of poisoning from grade 1 to grade 5, increased MDA levels (87.6% mean increase at grade 1 to 140% mean increase at grade 5)

(continued)

Table 2. Continued.

Authors, year	Study type	Sample collected ^a	Number of samples	Method	Results
Panda et al., 2015 [27]	Prospective study	Plasma	102 patients	Spectrophotometric method	MDA level at admission: 51.7 ± 6.1 nmol/ml Patient group: Mild: 48.5 ± 5.5 nmol/ml Moderate: 54.5 ± 6.4 nmol/ml Severe: 55.7 ± 6.7 nmol/ml Percentage difference: 2.3% mean increase from levels of MDA during admission Increase in severity of poisoning only from moderate to severe; decrease in MDA levels (6.1% mean decrease in patients with mild symptoms to 7.7% mean increase in patients with severe symptoms)
<i>Thiol groups (protein oxidation)</i>					
Dandapani et al., 2003 [30] ^b	Case-control study	Erythrocyte	19 patients; 5 healthy controls	Classical chemistry (protein thiols)	Control group: 0.06 ± 0.01 U/mg protein Patient group: 0.06 ± 0.02 U/mg protein (not statistically significant) Percentage difference: No mean difference at admission Intermediate syndrome (IMS) vs. non-IMS patients (statistically significant) IMS: 0.25 ± 1.13 U/mg protein Non-IMS: 0.21 ± 0.13 U/mg protein A mean increase of 19% was found in IMS patients compared to non-IMS patients
Ranjbar et al., 2005 [36]	Case-control study	Plasma	22 patients; unknown number of healthy controls	Spectrophotometric method (total thiol molecules)	Control group: 0.14 ± 0.01 mmol/L Patient group: 0.10 ± 0.01 mmol/L Percentage difference: a mean decrease of 28.6% was found in OP poisoned patients compared to control
Venkatesh et al., 2006 [28]	Case-control study	Erythrocyte	32 patients; 18 healthy controls	Classical chemistry (protein thiols)	Control group: 9.1 ± 4.45 U/mg protein Patient group: 22.59 ± 26.85 U/mg protein (not statistically significant) Percentage difference: 148.2% mean increase compared to control IMS vs. non-IMS patients (statistically significant) Day 7: Mean increase of 332.5% in patients with IMS compared to non-IMS Day 10: Mean increase of 509.7%
Soltaninejad et al., 2007 [33]	Case-control study	Plasma	21 patients; 26 healthy controls	Spectrophotometric method (total thiol molecules)	Control group: 0.85 ± 0.09 mmol/L Patient group: Mildly affected: 0.54 ± 0.06 mmol/L Severely affected: 0.4 ± 0.05 mmol/L Percentage difference: a mean decrease of 44.7% was found in OP poisoned patients compared to control
Superoxide dismutase (SOD) Banerjee et al., 1999 [29]	Case-control study	Erythrocyte	25–30 patients; unknown number of healthy controls	Classical chemistry	Control group: 1600 ± 200 units/g Hb Patient group: 3100 ± 125 units/g Hb Percentage difference: 93.75 mean increase was found among OP poisoned patients compared to control

(continued)

Table 2. Continued.

Authors, year	Study type	Sample collected ^a	Number of samples	Method	Results
Vidyasagar et al., 2004 [35]	Case-control study	Erythrocyte	84 patients; 10 healthy controls	Spectrophotometric method	Control group: 57.9 ± 6.9 IU Patient group: Grade 1: 71.5 ± 11.8 IU Grade 2: 74.7 ± 9.8 IU Grade 3: 86.2 ± 12.5 IU Grade 4: 86.4 ± 7.9 IU Grade 5: 94.2 ± 17.8 IU Percentage difference: 35.5% mean increase Increase in severity of poisoning from grade 1 to grade 5, increased SOD activity (23.5% mean increase at grade 1 to 62.7% mean increase at grade 5) No improvement in SOD activity with treatment (atropine and oximes) in 12 cases where both pre-treatment and post-treatment samples were collected Control group: 80.2 ± 119.5 U/ml Patient group: 59.7 ± 14.3 U/ml Percentage difference: A mean decrease of 25.6% was found when compared to control
Zhang et al. 2010 [32]	Case-control study	Serum	49 patients; 50 controls	Spectrophotometric method	Control group: 10.14 ± 3.6 units/g Hb Patient group: Grade 1: 13.82 ± 7.26 units/g Hb Grade 2: 13.19 ± 5.54 units/g Hb Grade 3: 14.39 ± 5.14 units/g Hb Grade 4: 15.75 ± 6 units/g Hb Grade 5: 17.84 ± 7.54 units/g Hb Percentage difference: A mean increase of 39.9% in SOD activity was found in OP poisoned patients compared to control Increase in severity of poisoning from grade 1 to grade 5, increased SOD activity (36.3% mean increase at grade 1 to 75.9% mean increase at grade 5)
Hundekari et al., 2013 [26]	Case-control study	Erythrocyte	150 patients; 30 healthy controls	Classical chemistry	Control group: 2.9 ± 0.8 units/g Hb Patient group: 3.6 ± 0.7 units/g Hb Percentage difference: A mean increase of 24.1% in CAT activity was found in OP poisoned patients compared to control Control group: 0.52 ± 0.42 U/mg protein Patient group: 0.22 ± 0.18 U/mg protein Percentage difference: mean decrease of 57.7% was found in poisoned patients compared to control group Control group: 15.54 ± 5.67 mg Hb/min Patient group: Grade 1: 18.44 ± 6.54 mg Hb/min Grade 2: 18.67 ± 7.03 mg Hb/min Grade 3: 19.25 ± 6.98 mg Hb/min Grade 4: 20.11 ± 7.9 mg Hb/min Grade 5: 21.28 ± 7.68 mg Hb/min Percentage difference: A mean increase of 21.5% in CAT activity was found in OP poisoned patients compared to control Increase in severity of poisoning from grade 1 to grade 5, increased CAT activity (18.7% mean increase at grade 1 to 36.9% mean increase at grade 5)
Catalase (CAT, antioxidant) Banerjee et al., 1999 [29]	Case-control study	Erythrocyte	25–30 patients; unknown number of healthy controls	Classical chemistry	
Venkatesh et al., 2006 [28]	Case-control study	Leucocytes	32 patients; 18 controls	Classical chemistry	
Hundekari et al 2013 [26]	Case-control study	Erythrocyte	150 patients; 30 healthy controls	Classical chemistry	

(continued)

Table 2. Continued.

Authors, year	Study type	Sample collected ^a	Number of samples	Method	Results
<i>Glutathione peroxidase (GPx, antioxidant)</i> Hundekari et al., 2013 [26]	Case-control study	Erythrocyte	150 patients; 30 healthy controls	Classical chemistry	Control group: 5351 ± 935 U/L Patient group: Grade 1: 6164 ± 1267 U/L Grade 2: 6035 ± 1210 U/L Grade 3: 6384 ± 1378 U/L Grade 4: 6007 ± 1355 U/L Grade 5: 6197 ± 1353 U/L Percentage difference: A mean increase of 12.5% in GPx activity was found in OP poisoned patients compared to control Control group: 5.1 ± 0.4 U/g Hb Patient group: 7 ± 0.25 U/g Hb Percentage difference: A mean increase of 37.2% in GPx activity was found in OP poisoned patients compared to control.
Banerjee et al., 1999 [29]	Case-control study	Erythrocyte	25–30 patients; unknown number of healthy controls	Classical chemistry	
<i>GSH reductase (GR, antioxidant)</i> Venkatesh et al., 2006 [28]	Case-control study	Leucocytes	32 patients; 18 controls	Classical chemistry	Control group: 37.86 ± 30.96 U/mg protein Patient group: 7.65 ± 6.79 U/mg protein Percentage difference: A mean decrease of 79.8% in GR activity was found in OP poisoned patients compared to control
Banerjee et al., 1999 [29]	Case-control study	Plasma	25–30 patients; unknown number of healthy controls	Classical chemistry	Control group: 0.9 ± 0.25 units/ml Patient group: 1 ± 0.5 units/ml Percentage difference: A mean increase of 11.1% in GR activity was found in OP poisoned patients compared to control

^aSamples collected at admission (except Dandapani et al.'s study).^bDandapani et al.'s study – samples were collected until discharge, therefore values used in the table refer to area under the curve of temporal profiles of mean levels of lipid peroxidation and protein oxidation.

Table 3. Antioxidant intervention studies in animals.

Study, year	Organophosphate and dosage	Drug and dosage	Number of animals	Outcome measures	Key results
Peña-Llopis et al., 2003 [25]	1.5 mg/L dichlorvos	1 mmol kg ⁻¹ acetylcysteine	30 eels – control group 100 randomly selected eels divided into two groups – 50 fish anaesthetised with ice were injected with acetylcysteine; the rest with saline	Mortality, biochemical parameters	- Eels pre-treated with 1 mmol kg⁻¹ of acetylcysteine presented a 66.7% higher survival than control eels. - Muscle GSH/GSSG (Mean ± SD): → Saline group control: 26.1 ± 2.1 nmol saline-treated: 12.5 ± 0.6 nmol → Acetylcysteine-treated group: 17.1 ± 0.9 nmol - GSH/GSSG ratio was reduced overall compared to controls, but percentage difference as higher by 12.5% in acetylcysteine-treated groups than saline-treated groups - Heart rate increased in acetylcysteine-treated group by nearly 50% at 120 mins. → Control: 121.75 ± 17.91 beats per minute → Acetylcysteine-treated: 182.25 ± 45.05 beats per minute - Increased SOD and CAT activity, and MDA levels in control group - No overall increase in survival rate
Cankayali et al., 2005 [22]	70 mg/kg dichlorvos	150 mg/kg acetylcysteine	15 rats Group 1 (n = 7) (control): (IP) 70 mg/kg dichlorvos+(IV) 0.9% NaCl (placebo) Group 2 (n = 8) (acetylcysteine-treated): (IP) 70 mg/kg dichlorvos+(IV) acetylcysteine (150 mg/kg)	MDA, SOD and CAT activity, survival rate, haemodynamic markers	- MDA levels were least in group 1 (1.17 ± 0.1 nmol/ml) and maximum in group 2 (1.91 ± 0.5 nmol/ml). Prophylactic (1.24 ± 0.3 nmol/ml) and therapeutic acetylcysteine (1.25 ± 0.1 nmol/ml) significantly decreased levels of MDA compared to control (group 2). - GSH levels were significant lower than in group 2 (29.87 ± 3.3 mg/dl) than group 1 (24.36 ± 3.1 mg/dl), 3 (28.87 ± 2.9 mg/dl), and 4 (31.49 ± 2.7 mg/dl) - Case fatality: Survival increased by 37.5%, 37.5% and 62.5% in acetylcysteine 50 mg/kg, 100 mg/kg and 200 mg/kg groups, respectively, within 6 h. The case fatality was highest in the control group with 12.5% surviving within first 6 h, and 100% survived in the atropine group
Yurumez, et al., 2007 [23]	100 mg/kg fenthion (for biochemical analysis)	150 mg/kg acetylcysteine	30 mice Group 1 (n = 6) – no agent; Group 2 (n = 8) – fenthion + saline; Group 3 (n = 8) – fenthion + prophylactic acetylcysteine Group 4 (n = 8) – fenthion + therapeutic acetylcysteine	MDA, GSH, nitrate, nitrite	- MDA levels were least in group 1 (1.17 ± 0.1 nmol/ml) and maximum in group 2 (1.91 ± 0.5 nmol/ml). Prophylactic (1.24 ± 0.3 nmol/ml) and therapeutic acetylcysteine (1.25 ± 0.1 nmol/ml) significantly decreased levels of MDA compared to control (group 2). - GSH levels were significant lower than in group 2 (29.87 ± 3.3 mg/dl) than group 1 (24.36 ± 3.1 mg/dl), 3 (28.87 ± 2.9 mg/dl), and 4 (31.49 ± 2.7 mg/dl) - Case fatality: Survival increased by 37.5%, 37.5% and 62.5% in acetylcysteine 50 mg/kg, 100 mg/kg and 200 mg/kg groups, respectively, within 6 h. The case fatality was highest in the control group with 12.5% surviving within first 6 h, and 100% survived in the atropine group
	200 mg/kg fenthion (for mortality study)	50, 100 and 200 mg/kg acetylcysteine intraperitoneal (i.p.) 1 h after fenthion administration (compared with atropine and control)	40 mice for mortality study Group A (n = 8) – fenthion + saline; Group B (n = 8) – fenthion + atropine Groups C, D, E (n = 8 each) – fenthion i.p.+acetylcysteine 50, 100, 200 mg/kg 1 h later, respectively	Case fatality	

(continued)

Table 3. Continued.

Study, year	Organophosphate and dosage	Drug and dosage	Number of animals	Outcome measures	Key results
Bayir et al., 2013 [24]	50 mg/kg dichlorvos	Coenzyme Q10 (CoQ10) as an adjunct 50 mg 0.5 mg/kg IV atropine 15 mg/kg IV PAM every 4h; after 30 mg/kg IV bolus	20 rabbits Control (n = 8), PAM + atropine (n = 6), CoQ10 + PAM + atropine (n = 6)	Nitric oxide and MDA	- Mean erythrocyte higher in CoQ10 group than other two groups at 12 h > atropine group at 24 h - Mean erythrocyte nitric oxide levels at 12 (4.1 mmol/g Hb) and 24 h (4.9 mmol/g Hb) after poisoning in CoQ10 + PAM + atropine group were significantly lower than PAM + atropine group (6.1 mmol/g Hb at 12 h and 7.9 mmol/g Hb at 24 h) - Mean MDA levels were lower in CoQ10 + PAM + atropine group (5.7 nmol/ml at 12 h and 6 nmol/ml at 24 h) than control (9 nmol/ml at 12 h) and PAM + atropine group (8.9 nmol/ml at 12 h and 9.2 nmol/ml at 24 h)

intermediate syndrome patients, which correlated with duration of ventilatory failure and severity of poisoning [30].

DNA damage

Only one small study ($n = 13$ patients) measured oxidative DNA damage after acute OP poisoning [31]. The study found that 8-hydroxydeoxyguanosine levels were lower in OP poisoned patients compared to control, showing no positive association between oxidative DNA damage and acute OP insecticide poisoning [31].

Nitrosative stress

Two case-control studies investigated nitrosative stress in acute OP insecticide self-poisoning [33,34]. One study measured nitrite as a surrogate of nitric oxide spectrophotometrically and found that increasing plasma nitrite correlated with decreasing acetylcholinesterase activity [34]. A second that measured plasma nitrite concentration quantitatively, using a classical chemistry method, did not find a significant increase in OP insecticide poisoned patients compared to control groups [33]. The latter negative result is unsurprising because the threshold for detection of nitrite by the Griess test is $\sim 1 \mu\text{M}$, whereas the concentrations detected in this study were $\sim 60 \text{ nM}$. The Griess test is not appropriately sensitive for use in plasma samples – chemiluminescence testing is required and the samples need to be very carefully handled post-sampling to avoid auto-oxidation of the target nitrite to nitrate.

Antioxidant defence system

Six studies [26,28,29,32,35,36] measured antioxidants, antioxidant-dependent enzymes, or total antioxidant capacity to record changes in the antioxidant defence system in OP poisoned patients.

Antioxidants and antioxidant-dependent enzymes

Five studies measured one or more of the following antioxidants or antioxidant-dependent enzymes – superoxide dismutase, catalase, glutathione, glutathione peroxidase, glutathione reductase and paraoxonase 1 – to assess antioxidant activity in acute OP insecticide poisoning [26,28,29,32,35]. Three of five studies found that antioxidant activity significantly increased in early acute OP poisoning, although the method(s) used have serious limitations.

Of four studies assessing superoxide dismutase, three studies found an overall mean increase of 56.4% (range: 35.5% to 93.75%) in erythrocyte superoxide dismutase activity (Figure 3) [26,29,32,35]. However, a fourth measured serum superoxide dismutase and found it to be 25.6% lower in poisoned patients compared to controls [32]. Two studies found an increase in erythrocyte superoxide dismutase activity that was related to clinical severity [26,35]. Treatment with atropine and oximes had no measurable effect on superoxide dismutase activity [35].

Three case-control studies measured the activity of catalase [26,28,29]. Two found increases of 24.1% and 21.5% [26,29]. The exception found it to be decreased by more

Table 4. Antioxidant intervention studies in OP poisoned patients.

Study, year	Drug and dosage	Number of patients	Outcome measures	Results
Shadnia et al., 2011 [7]	Acetylcysteine (IV) 140 mg/kg as a loading dose; 70 mg/kg/4h for 17 doses	24 patients were randomly allocated to two groups – conventional treatment with acetylcysteine ($n = 12$) vs. conventional treatment only ($n = 12$)	Case fatality, total dose of atropine and pralidoxime administered, duration of hospitalisation	Acetylcysteine treatment decreased duration of hospitalisation by 49% and lowered required atropine doses by 76.6% Case fatality was reduced in acetylcysteine treated group (1/12 case in acetylcysteine treated and 3/12 cases in control), however, it was not statistically significant
El-Ebiary et al., 2016 [8]	Acetylcysteine (oral) 600 mg TDS for 3 days	30 patients were randomly allocated two groups – conventional treatment plus acetylcysteine ($n = 15$) vs. conventional treatment plus placebo ($n = 15$)	Case fatality, total dose of atropine administered, duration of hospitalisation and need for ICU admission and/or mechanical ventilation	Acetylcysteine treatment lowered required atropine doses by 54.5% MDA levels significantly decreased by 52.3% and reduced glutathione increased by 338.8% post-treatment in groups treated with acetylcysteine No significant difference in duration of hospitalisation and no deaths were reported

than 50% in leucocytes of acutely OP insecticide poisoned patients, before increasing to 137% and 220% of baseline in patients with intermediate syndrome at day 7 and 10 of hospitalisation, respectively [28].

Two studies measured erythrocyte glutathione peroxidase activity; both found a mean increase of 25% (range: 12.5% to 37.2%) in poisoned individuals compared to control [26,29]. Two small studies measured glutathione reductase with varied results: one study found glutathione reductase activity to be reduced by nearly 80% in patients compared to controls, while the second reported a mean increase of 11% in patients compared to controls but with high individual patient variability [28,29]. Up-regulation of glutathione reductase might be the expected adaptive response to oxidative stress-induced increased glutathione oxidation.

Total antioxidant capacity (TAC)

Two studies found the total antioxidant capacity (measured by the non-recommended plasma ferric reducing ability assay [Box 2]) was decreased [26,36], and correlated with clinical severity in one study [26]. However, measuring total antioxidant capacity in plasma may not reflect the antioxidant status, making it a poor marker of oxidative stress. The assay itself strictly measures reducing capacity rather than ability to scavenge reactive oxygen species: while the two chemical characteristics are not unrelated, the ability of a sample to reduce Fe^{3+} to Fe^{2+} lacks relevance. One study that measured total antioxidant capacity along with antioxidants superoxide dismutase and catalase found a significant increase in activity of superoxide dismutase and catalase but a reduction in total antioxidant capacity [26]. In another study, an overall decrease in both total antioxidant capacity and total plasma thiols was noted in 22 patients [36].

Free radical generator

Two studies that directly measured xanthine oxidase (which generates free radicals, Figure 1) showed a two- to five-fold increase in activity in patients [28,32].

Antioxidant intervention studies in animals

We found four animal studies that tested antioxidant medicines for acute OP poisoning (three using acetylcysteine and one using coenzyme Q10) (Table 3) [22–25].

Studies were at risk of bias as per SYRCLE's risk of bias tool (Figure 4). One small study showed that acetylcysteine 200 mg/kg given intraperitoneally (i.p.) 1 h after 0.2 g/kg fenthion i.p. increased survival (from 12.5% with control to 62.5% with acetylcysteine), when compared with controls and atropine-treated groups [23].

Two studies showed improvement in oxidative stress biomarkers in rabbits and fish poisoned with dichlorvos and treated with coenzyme Q10 and acetylcysteine, respectively [24,25]. The fourth study showed that acutely poisoned rats treated with acetylcysteine had a survival rate of 100% in the first 60 min compared to control groups which had a survival rate of 85%; however, the survival rate at 120 min were 57% and 50% in control and acetylcysteine-treated groups, respectively [22]. The statistical significance for survival rate and methodology for how it was measured is not known for this study.

Antioxidant intervention studies in humans

We found only two small randomised control trials (total patient number = 54) considering antioxidant treatments in human acute OP insecticide self-poisoning, both of which tested acetylcysteine in addition to standard atropine and oxime therapy [7,8] (Table 4). Both studies were very small,

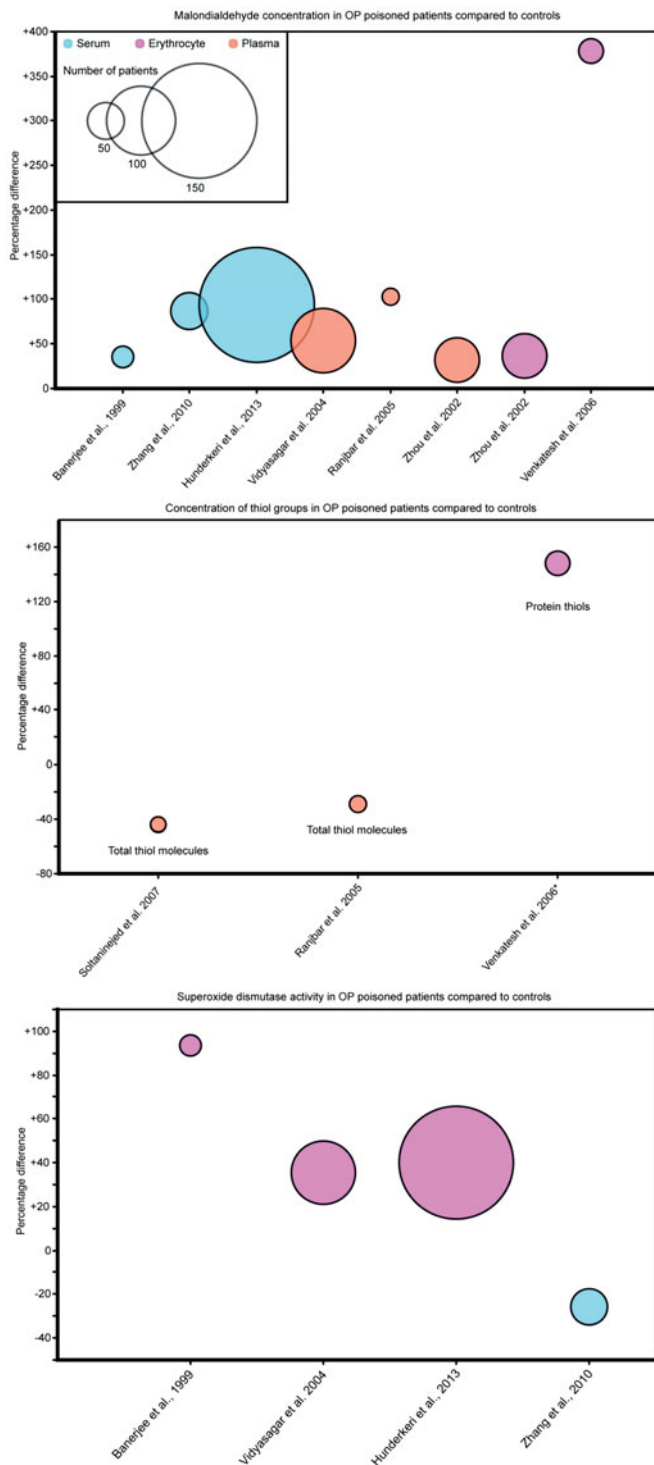


Figure 3. Change in malondialdehyde concentration, concentration of thiol groups, and superoxide dismutase activity compared to control in OP poisoned individuals. * denotes results that were not found to be statistically significant.

without any power calculation, and insufficiently powered to show even a modest effect [39]. They were also at risk of bias as per Cochrane collaboration guidelines (Figure 5).

Both RCTs found that acetylcysteine reduced atropine dosage requirements by 76.6% (control group: 38.9 ± 28.8 mg; patient group: 9.1 ± 7.2 mg) and 54.5% (control group: 11 ± 2 mg; patient group: 5 ± 4 mg) without any statistically significant reduction in case fatality or other secondary outcomes [7,8].

Study	Sequence generation	Baseline characteristics	Allocation concealment	Random housing	Blinding of caregivers and/or investigators	Random outcome assessment	Blinding of outcome assessment	Incomplete outcome data	Selective outcome reporting	Other sources of bias
Cankayali et al., 2005 [22]	?	+	?	+	-	+	?	-	+	-
Yurumez, et al., 2007 [23]	?	?	?	?	-	+	?	+	+	+
Bayir et al., 2013 [24]	?	?	?	?	?	?	?	+	+	-
Peña-Llopis et al., 2003 [25]	N/A	+	N/A	+	?	+	?	-	-	-

Figure 4. Assessment of potential bias using SYRCLE's risk of bias tool for animal studies. + low risk of bias, - high risk of bias; ? unclear risk of bias. Other bias includes small sample size, the variable effect of OP in different animals, and potential contamination. + low risk of bias, - high risk of bias; ? unclear risk of bias.

Discussion

We found multiple studies addressing the occurrence and importance of oxidative stress in acute OP insecticide self-poisoning. A consistent rise in endogenous antioxidant capacity was found in several studies we reviewed [26,29,35], but the techniques used were weak and this result is open to interpretation with respect to oxidative stress. Many reported changes in oxidative damage biomarkers, suggestive of oxidative stress in poisoned patients that was not totally countered by endogenous antioxidants. However, the quality and reliability of many of the assays used in the studies were not ideal. In addition, no account was taken of the particular OP insecticides studied, there was no consistent assessment of poisoning severity, and the time point in the clinical course of blood sampling was inadequately described, making comparisons between studies very difficult. Overall, the current data are insufficient to determine whether the increase in antioxidant defences is able to protect proteins, lipids, and DNA from oxidative damage in acute OP insecticide self-poisoning.

Pre-clinical rodent studies suggested that antioxidants acetylcysteine and coenzyme Q10 might improve survival and reduce the clinical complications of acute OP poisoning; however, studies with low risk of bias were few. Given that the outcomes measured were survival, the acetylcysteine studies might simply indicate increased regeneration of hepatic glutathione to help with detoxification of the OP, rather than an antioxidant activity *per se*.

Only two small randomised control trials of antioxidants in poisoned patients have been reported [7,8]; while showing a reduction in the need for atropine, these studies failed to measure any impact on clinically relevant outcomes. They were also too small to show even a modest effect and were at risk of bias. Far more clinical data are required before antioxidant medicines can be recommended for routine clinical use in OP poisoning.

Study	Random sequence generation	Allocation concealment	Blinding of study participants and personnel	Blinding of outcome assessment	Incomplete outcome reporting	Selective outcome reporting	Other biases
Shadnia et al., 2011 [7]	?	?	+	-	+	-	-
El-Ebiary et al., 2016 [8]	+	+	+	?	+	-	-

Figure 5. Assessment of potential bias using Cochrane collaboration guidelines for human studies. Other bias includes small sample size and lack of identification of OP pesticide. + low risk of bias, - high risk of bias; ? unclear risk of bias.

Looking at particular elements of oxidative stress, the data on lipid peroxidation are inconclusive. While the concentrations of malondialdehyde and conjugated dienes show consistent mean increases, we note that there are both large standard deviations within studies themselves, and significant variability between studies. Furthermore, although malondialdehyde has regularly been used as a marker of oxidative stress, it lacks specificity for measuring lipid peroxidation and measures of more reliable markers are necessary to reach firm conclusions. These studies did not investigate markers of oxidative stress over several hours after poisoning that might have explained the variation as being due to different times of sampling after poisoning.

Evidence that oxidative protein damage occurs in acutely poisoned patients is similarly inconclusive. Two studies found an overall decrease in chemically reduced total thiol concentrations [33,36], suggesting no oxidative stress, while other studies found an increase or no change in reduced protein thiol groups on admission [28,30,33,36]. Reduced total thiol measurements were elevated in intermediate syndrome patients after 7 days. Two studies suggest that protein and lipid oxidative damage is associated with intermediate syndrome [28,30], although further studies are required to assess whether the relationship is real and causal. Increase in clinical severity and hospital stay were associated with changes in protein damage markers, suggesting that other factors such as treatment regimes, hypoxia or immobilisation could affect the oxidation state of thiol group concentrations.

There are several studies which suggest that nitric oxide plays a role in oxidative damage during acute OP poisoning, but inappropriate methods have been applied and the outcomes are therefore unreliable. The presence of reduced antioxidant capacity in the absence of altered plasma nitrite concentration suggests that reactive oxygen species may play a more significant role in acute OP poisoning than reactive nitrogen species. However, rat studies examining nitrosative stress in skeletal muscle noted that monocrotophos poisoning inhibited mitochondrial complex I and activated mitochondrial complex II, while causing oxidative stress and nitrosative stress markers to rise in skeletal muscle [40]. However, oxidative damage was not detected [40].

More evidence suggests that endogenous antioxidants are activated rather than inhibited in acute OP poisoning. Chronic exposure to acute OP insecticides (for ≥ 6 months) increases antioxidant activity and DNA damage “in the absence of depressed acetylcholinesterase levels” [41]. For oxidative stress to cause oxidative damage, there must be a significant inhibition of antioxidant defences or a significant rise in reactive oxygen species that overcomes the adaptive rise in antioxidant activity. Since it is difficult to measure reactive oxygen species directly, it is difficult to prove or disprove that the rise in antioxidant activity is associated with oxidative stress. It is possible that the antioxidants are produced in an attempt to counteract reactive oxygen species that have been released as a result of the poisoning. Oxidative damage in the presence of increased antioxidant activity perhaps suggests that endogenous antioxidants are insufficient to protect macromolecules from being damaged.

Several animal studies reported improved outcomes with the use of acetylcysteine, a prodrug to L-cysteine, which is a precursor to the abundant intracellular antioxidant glutathione [19]. Glutathione plays a role in inactivating toxic metabolites (as in the case of treating acetaminophen overdose) [19]. Evidence suggests that acetylcysteine has poor antioxidant capacity in the plasma compartment and works by replenishing glutathione stores only in those with depletion. Hence, the improved outcomes reported by animal studies could be due to the detoxifying properties of *de novo* synthesised glutathione [42], rather than the antioxidant effects of either glutathione or acetylcysteine itself. The studies were few and lacked generalisability; in addition, the animal studies were not reflective of an episode of acute OP self-poisoning in humans.

Limitations of methodology of published studies

Sampling points

Seven out of ten human studies obtained a single or few blood samples, usually a sample soon after admission (pre-treatment sample) and/or at discharge (post-treatment sample). Two studies had longer observation periods and collected samples at different times. In the study conducted by Dandapani et al. [30] blood samples were collected on day 1 of admission and alternate days until discharge. In the Venkatesh et al.'s [28] study, samples were taken at admission and “on most days” for the first 10 days of hospitalisation. Zhou et al. [34] also took multiple samples but the publication does not specify when samples were collected.

Choosing an oxidative stress marker to measure

Thiobarbituric acid assay-based spectrophotometric methods or basic laboratory measurements of malondialdehyde can estimate its concentration in the blood, but do not provide information about the extent to which oxidative damage is associated with an increase in malondialdehyde. Oxidative protein damage is a better marker for oxidative stress; however, very few studies chose to measure oxidised proteins (e.g., protein carbonyls). In addition, the surrogate markers of

nitric oxide used in the studies were weak; chemiluminescence is by far the best detection method for nitrite in plasma. Nitrate is not a good indicator of endogenous nitric oxide generation because plasma levels are heavily influenced by dietary nitrate.

Clinical severity

Researchers used different scales to assess clinical severity based on symptoms. For instance, different studies used the Namba scale [43], the Peradeniya Organophosphorus Poisoning (POP) scale [44], that of Balany and Moses [35], or their own judgement. Furthermore, in most studies, very little information regarding the clinical course of acutely poisoned patients is given. This makes direct comparisons of clinical severity and therefore results between studies challenging.

Fluctuations in antioxidant activity

The variability between studies may be due to variations in time after poisoning. Rosalovsky et al. [45] have shown variations in the inhibition of antioxidant systems at different times after poisoning. The antioxidant capacity was reduced at 30 min and 45 min after an initial increase at 15 min suggesting that there may be a failure of the antioxidant system to maintain itself [45]. However, as rodent models have a higher metabolic clearance of OP compounds, the late reductions in antioxidant system could be due to clearance of OP insecticide.

Conclusions

We found multiple studies showing raised antioxidant activity after acute OP poisoning, suggesting that protective mechanisms are activated. However, the evidence of damage to proteins and lipids is inconclusive, and it is unclear whether oxidative damage occurs in humans with acute OP poisoning. Existing animal and human studies provide weak data, which is insufficient to draw conclusions regarding the role of oxidative stress in causing clinical harm after acute OP insecticide poisoning. Although its role is increasingly being investigated, many studies lack value due to small sample size and issues with the assays chosen to measure oxidative stress. More evidence in the form of large high-quality, prospective randomised control trials with oxidative stress markers measured before and after treatment with antioxidants are required to determine whether antioxidants might be beneficial in the management of acute OP insecticide self-poisoning.

Disclosure statement

No potential conflict of interest was reported by the authors.

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